Erica Lopez

zznailmail@gmail.com · New York City Metropolitan Area, NY

https://www.linkedin.com/in/ericcagal

https://github.com/lllcloudstudio

Summary

Expertise in R statistical programming, object oriented programming Python and SQL to transform complex manual processes into high-performance software experiences, specializing in fullstack development. 9+ years of research experience and a proven track record of published projects (7 Google Scholar) papers. Designed a collaborative data acquisition analysis and governance tool for university hospital lab team(s) at Tufts Medical Center (working on either windows or Mac systems); complementing proteomics or mass spectrometry research software together with 500+ separate data samples.

Machine Learning Assessment Summary

See the data acquisition analysis and governance tool and a machine learning report with yellowbrick at *https://youtu.be/NPYj2m8MKYA and https://youtu.be/zj9TQT4qKTU, respectively. Code reference at github.*

A 12+ criteria review of 2017 health data at Boston University from Dr. Vijay Kolachalama, PhD. Data is either synthetic or not.

50 patient blood sample(s) 13 diagnostic variables, numerics and without annotations. 3 categorical variables are Age, Gender, or Race. The sample data is considered a small sample size, less than 300 total samples (<300*)

Variables: DBP, SBP, ph, LDL, HDL, or hb are diagnostics, I consider an i) extrapolation to modelling towards review (by pearson or spearman) and selection for further assessment and reduction for data review ii) indicator(s) of health status, or iii) variance explanation by PCA.

The two group (Male or Female) health bias is delved machine learning (ML) with either k-means, support vector machine (svm), or hierarchical clustering (hc) for: model selection (dimension reduction) by number of features.

Review:

1. Summary comparison of 2 models (using the R glm() function) either category or Gender as response.
2. PCA (R prcomp()) function is centered and scaled principal component score vectors have length n=50 and the principal component loading vectors have length p=14

3. Biplot displays 2 components or "loading vectors" listed at the table with colnames PC1, or PC2, etc. Summary per component: Standard deviation, Proportion of Variance, or Cumulative Proportion multiplot line or bar graph of vectors.
4. K-means semi-step wise evaluation to k=6, and 1 additional parameter.
5. Model evaluation: training and test set selection (pairing 10-40 or pairing 15-35) LOOCV* or k-fold cross validation*.

Inspection of data by category is a preliminary procedure to model(s) production (reducing the number of models), evaluation, or selection.

Work Experience

Associate Research Fellow *May 2025 – Present*
THE CITY COLLEGE OF NEW YORK, NEW YORK, UNITED STATES

- Design and implement Python-based automated extraction pipelines to transform raw PDB files into structured datasets for water-to-water interaction analysis.
- Transform manual distance assessments into automated API-driven scripts, reducing data processing bottlenecks by 40%.
- Ship high-quality code utilizing R and Python3 to evaluate functional water networks, ensuring strict compliance with peer-reviewed literature standards.
- Implement Postgres-based data storage solutions to manage large-scale molecular datasets, optimizing query performance for complex hydrogen bond interactions.

Research Associate*October 2020 – November 2025*
TUFTS MEDICAL CENTER, BOSTON, MA

- Engineered a Python3 shell-compatible script to automate XML file generation, accelerating MaxQuant executable program compatibility and reducing manual setup time by 50%.
- Designed and implemented robust Python and R scripts to generate mathematical summaries of patient data, improving reporting accuracy for compliance workflows.
- Transformed disparate patient sample information into a centralized data architecture to support large-scale proteomics research.
- Shipped technical documentation and automated workflows that streamlined data exchange between lab systems and bioinformatics platforms.

Intern*May 2019 – August 2019*
BRIGHAM AND WOMEN'S HOSPITAL, BOSTON, MA

- Executed non-parametric statistical assessments on quantitative datasets, implementing automated modeling for Multiple Sclerosis patient samples.
- Quantified differential expression from RNA-seq datasets, transforming raw genomic data into actionable insights for therapeutic development.
- Designed data validation scripts to ensure compliance with clinical research protocols and data integrity standards.
- Implemented data cleaning micro-services to prep RNA datasets for high-throughput sequencing pipelines.

Intern*May 2018 – August 2018*
BOSTON CHILDREN'S HOSPITAL, BOSTON, MA

- Accessed Gene Expression Omnibus APIs to retrieve phenotype and microarray data tables, automating the ingestion of vaccination study datasets.
- Implemented Principal Component Analysis (PCA) to evaluate microarray expression probes and transform sample clustering visualizations.
- Designed system architecture for phenotype data tracking, improving the efficiency of cross-study data comparisons.
- Shipped modular analysis scripts that allowed for rapid iteration of yellow fever vaccine data assessments.

Postbaccalaureate IRTA*February 2015 – July 2017*
NATIONAL INSTITUTES OF HEALTH (NIH): INTRAMURAL RESEARCH PROGRAM (IRP), BETHESDA, MD, UNITED STATES

- Applied drug inhibition and knockdown methodologies to quantify complex gene expression patterns in long non-coding RNA.
- Identified transcript suppressors to transform scientific understanding of gypsy insulator complex localization within cell molecular biology.
- Implemented rigorous data collection protocols to maintain laboratory compliance and experiment reproducibility.
- Designed experimental workflows that automated the quantification of Flamenco gene expression in D. melanogaster cells.

ORISE Fellow*June 2014 – February 2015*
FDA, SILVER SPRING, MD

- Incorporated viral antigens into a live attenuated rubella vector, implementing innovative biotechnology techniques for vaccine development.
- Tested for rubella structural protein expression using western blot analysis, transforming raw cell lysate data into definitive antigenic insert results.
- Managed laboratory infrastructure and equipment to support viral vector research and high-volume antigenic testing.
- Designed testing protocols that ensured compliance with FDA safety and efficacy standards for attenuated vectors.

Education

Master's degree in College of Engineering BioinformaticsJuly 2017 – December 2020
Boston University, Boston, MA

Bachelor's degree
University of Connecticut

Additional Skills

Fullstack Development: React, TypeScript, Node.js, Python Object Oriented Programming, Fullstack Development, Java Native Interface Usage, API Design, JSON Website Applications

Data & Architecture: Postgres, SQL Warehouse Management, System Architecture, Google Cloud Platform (GCP), Infrastructure Development, Nextflow Pipeline Flow, XML GUI Development

Analysis & Compliance: Compliance, Permitting Workflows, Construction Technology, Machine Learning to Modeling, Support Vector Machine, K-Means, K-NN Machine Learning, Principal Component Analysis, Hierarchical Clustering

Tools & Environment: Git, UNIX Bash Shell, HTML Website Design, CSS, Cross-fold Validation, R Statistical Assessment

Languages

- Spanish - Native or Bilingual

Projects

Gunner Lab
A waters assessment, validation documents from pdb containing chain IDs of residues included in the properties of each water 9 molecule, using Schrodinger's pymol viewer or python3 to extract to excel.

execute-maxquant-non-gui

A python3 —shell compatible—script which generates an XML file containing specified parameters, compatible with the Max Planck MaxQuant executable program.

Publications

- Activation of PKR by short stem– loop RNAs containing single-stranded arms.
- Cross-platform proteomics signatures of extreme old age
- Robust Serum Proteomic Signatures of APOE2
- A Robust Serum Proteomic Signature of the E2 Allele of Apolipoprotein E
- Isha is a su(Hw) mRNA-binding protein required for gypsy insulator function